Analyzing Math Major Persistence

Using Logistic Regression

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SUU Festival of Excellence · March 2026 · Faculty Mentor: Dr. Bahi

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Background & Motivation

Framing persistence and early departure in the mathematics major

THE PROBLEM

Prior research suggests that the mathematics major experiences relatively high switching rates within STEM.

BREAKING IT DOWN FURTHER

Switching is typically analyzed as a single outcome,
which may obscure differences between early departure and longer-term persistence.

THIS STUDY

This study uses seven years of student data from a public university.

All predictors are measured at entry into the mathematics major.

Two questions guide the analysis: overall persistence and *switching after the first term*.

Research Questions

Two complementary views of persistence in the mathematics major

RQ1 — Long-Term Persistence

Which academic and demographic characteristics distinguish students who persist in the mathematics major from those who switch?

Examines the full observation window from entry to outcome

Outcome: Persisted vs. Switched

RQ2 — Immediate Switching

Which entry-level characteristics are associated with switching after the first term?

Zooms in on the earliest exit point: switching after one term

Outcome: One-term switcher vs. Persisted

Data & Method

Dataset

Item	Detail
Source	Institutional administrative records
Time span	Seven academic years of longitudinal student data
Population	Undergraduates who declared math Education or math sciences
Sample size	N = 170 students (one record per student)

Analytic Approach

Logistic regression in R

Binary outcome — persist vs. switch

Model selection

AIC comparison across specifications

Missing data

Predictive Mean Matching via mice (R)

Temporal validity

All predictors measured at entry only

Variables & Model

From entry-level predictors to the final parsimonious model

OUTCOME

Persistence in Mathematics Major

1 = persisted
0 = switched

OR = Odds Ratio

INITIAL VARIABLES

- Age at entry
- Gender
- Marital status
- Transfer status
- Financial aid status
- Enrollment status
- Double major status
- Math ACT score
- First-term GPA
- Math credit hours

→

FINAL MODEL

First-Term GPA
OR = 1.96 **

Math Credit Hours
OR = 1.19 **

ACT Math Score
OR = 1.11 ◦

*Selected by AIC (204.72) · ** p < .01 · ◦ p = .053*

$$\text{logit}(P) = \beta_0 + \beta_1(\text{First-term GPA}) + \beta_2(\text{Math credit hours}) + \beta_3(\text{Math ACT score})$$

Sample Overview — Outcome Distribution

N = 170 students across seven academic years

64

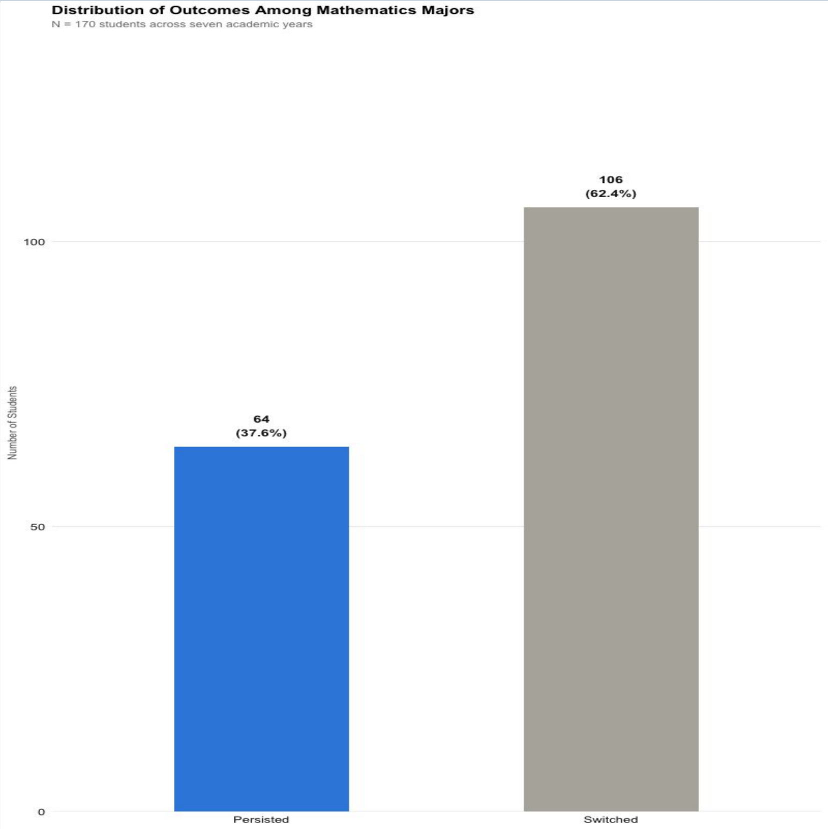
persisted in the mathematics major

37.6%

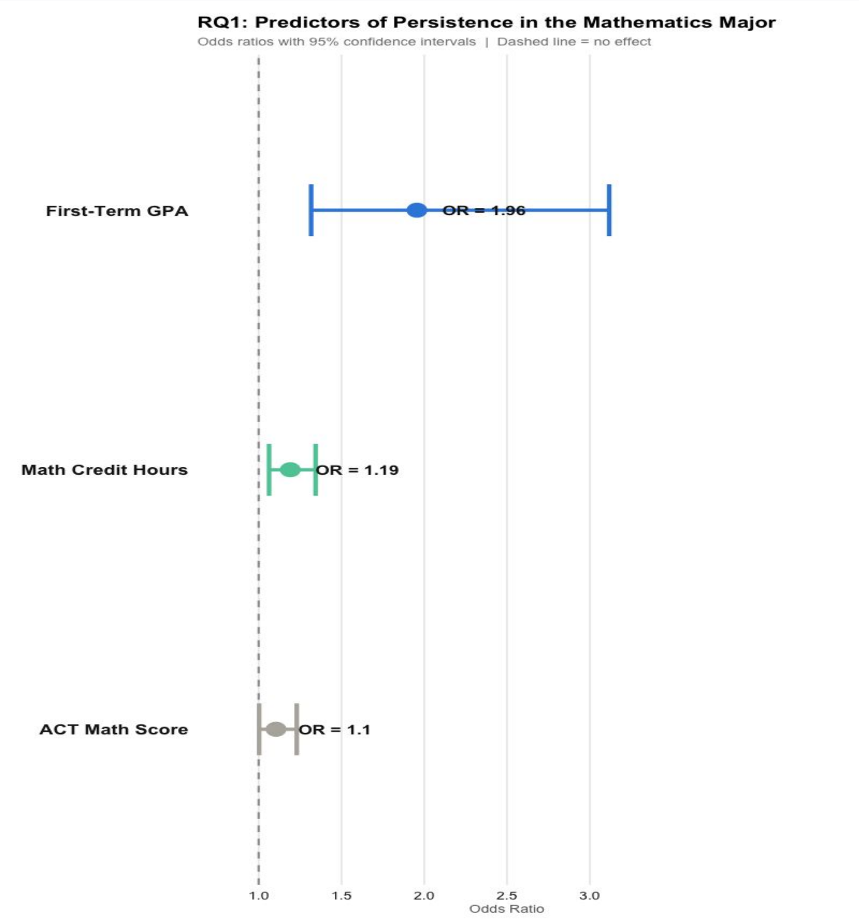
106

switched out of the mathematics major

62.4%



RQ1 Findings — Predictors of Long-Term Persistence



Interpretation

OR= 1.96 | 95% CI: [1.32, 3.12]

First-Term GPA

Each one-point increase in GPA is associated with a 96% increase in the odds of persistence.

OR = 1.19 | 95% CI: [1.06, 1.34]

Math Credit Hours

Each additional math credit hour is associated with a 19% increase in the odds of persistence.

OR = 1.11 | 95% CI: [1.00, 1.23]

ACT Math Score

confidence interval just exceeds 1.0

Best-fit model selected by AIC · AIC = 204.72 · Hosmer-Lemeshow p = 0.165

Model Building — What Else Was Tested?

Variables tested but not retained in the final model

Variables Tested

- Age at entry
- Gender
- Marital status
- State residency
- Transfer status
- Entry action
- Enrollment status
- Financial aid
- Double major status
- Attempted credit hours

What We Found

Demographic and financial variables were not significant when included with academic variables.

Financial aid was significant as a single predictor but not after including academic variables.

Academic variables remained strongest in the combined model.

Final Model Retains

First-Term GPA (OR 1.96 **)

Math Credit Hours (OR 1.19 **)

ACT Math Score (OR 1.11 ◦)

Selected by AIC · Parsimonious model preferred

$$\text{logit}(\text{Persistence}) = \beta_0 + \beta_1(\text{Term GPA}) + \beta_2(\text{Math Hours}) + \beta_3(\text{ACT Math})$$

RQ2 — Timing of Switching Among Mathematics Major Leavers

N = 170 students across seven academic years

46

switched after their first term

43.4%

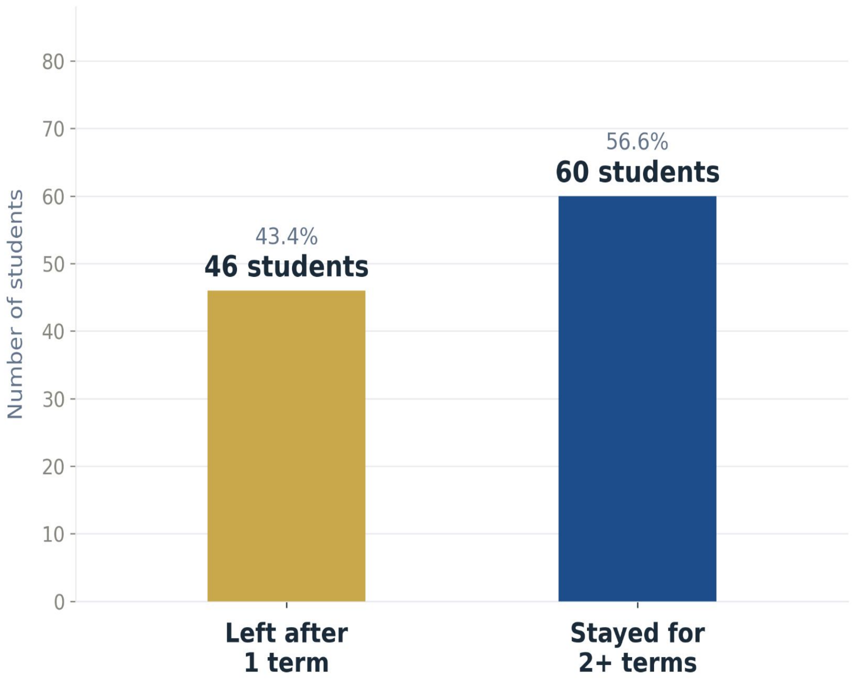
60

switched after more than one term

56.6%

Switching After One Term vs. Later Among Mathematics Major Leavers

Among the 106 students who switched out of the mathematics major



RQ2 Findings — Immediate Switching

Which entry-level characteristics are associated with switching after the first term?

Findings

- Entry-level variables tested included age at entry, ACT Math score, gender, marital status, transfer status, entry action, and financial aid (all measured at entry before switching).
- In this dataset, no entry-level variables were significantly associated with immediate switching.
- Neither academic, demographic, nor financial variables were significantly associated with immediate switching.

Interpretation

- Within this dataset, no association was observed between entry-level variables and immediate switching.
- This suggests that early departure may be influenced by factors not captured at entry.
- Despite this, a substantial share of students leave the major after only one term.

The Central Finding

Two models — two different stories about who leaves and why

Predictor	RQ1 — Long-Term Persistence	RQ2 — Immediate Switching
First-Term GPA	**Strong ✓ (OR = 1.96, $p < .01$)**	N/A
Math Credit Hours	**Strong ✓ (OR = 1.19, $p < .01$)**	N/A
ACT Math Score	Marginal ◦ (OR = 1.11)	Not significant
Demographic & Financial Variables	Not significant	Not significant

RQ1: Academic variables were associated with long-term persistence. RQ2: No entry-level variables were associated with immediate switching in this dataset.

Limitations

Scope, generalizability, and data constraints

Small sample

N = 170 limits statistical power — especially for detecting effects in less common subgroups such as specific demographic categories.

Single modeling approach

This study uses logistic regression only. Other methods may capture different patterns in the data.

Observational design

Results reflect associations, not causal relationships.

Conclusions & Implications

What the data show

First-term GPA and math credit hours are **predictors of long-term persistence**

ACT Math score shows shows weak evidence as a predictor ($p = .053$).

Entry-level variables were **not predictors of immediate switching**

A substantial share of students leave the major after only one term.

Retention is not one problem — it is at least two. Solving it starts with knowing which one you are facing.

Thank You

Questions & Discussion

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